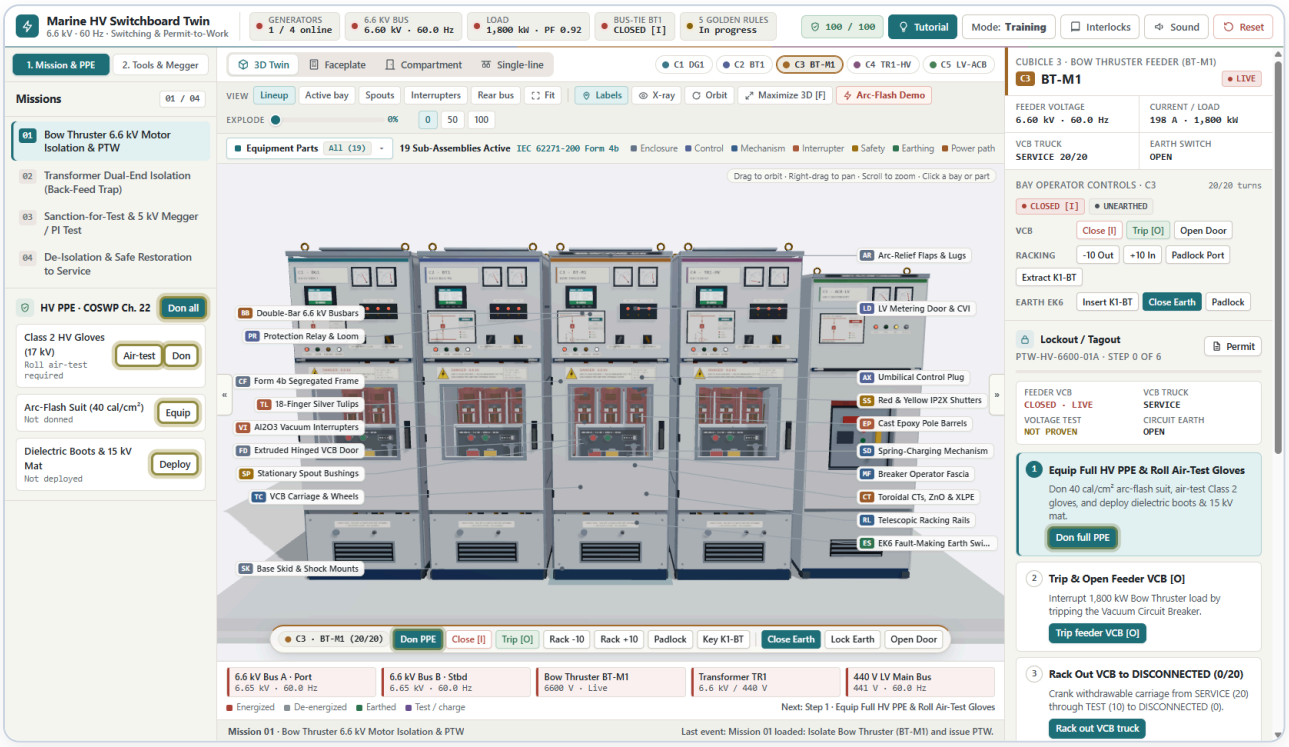
STCW 2010 MANILA AMENDMENTS | HIGH-VOLTAGE SAFETY (6.6 KV / 440 V) | IEC 62271-200 FORM 4B

COSWP CHAPTER 16 COMPLIANT

Marine High-Voltage 6.6 kV Switchboard Simulator

Comprehensive Operator's Guide, 3D Digital Twin Navigation Manual, and STCW Lockout-Tagout (LOTO) Switching Procedures



Training Directive & Purpose: This document provides a complete visual and operational guide for marine engineers, electrical officers (ETO), and technical crew operating marine high-voltage (HV) electrical propulsion and distribution plants. It details the navigation of the 3D switchboard digital twin, cubicle anatomy, 19 sub-assemblies, vacuum arc interruption kinematics, and STCW/COSWP safe isolation protocols.

AUTHOR / ORGANIZATION

Marine Engineering Simulation Group

SWITCHGEAR RATING

6.6 kV · 1250 A · 40 kA 1s · Form 4b

DOCUMENT RELEASE

Edition 2.2 · Verified Clean Build

1. Executive Summary & Electrical Architecture

Marine electrical power systems on modern vessels—including cruise liners, dynamic positioning (DP) offshore drillships, shuttle tankers, and container carriers—increasingly utilize High-Voltage (HV) distribution operating at **6.6 kV AC, 60 Hz**. High voltage is required to deliver megawatt-scale power to electric propulsion thrusters and heavy engine-room machinery while minimizing conductor cross-section, cable weight, copper losses, and short-circuit fault stresses.

However, 6.6 kV power introduces extreme safety hazards, including catastrophic arc-flash blasts (temperatures exceeding 19,000 °C), lethal shock hazards, and severe transient overvoltages. The **Marine High-Voltage 6.6 kV Switchboard Simulator** delivers a high-fidelity, interactive 3D digital twin of a marine main switchboard (MSB) built to **IEC 62271-200 Form 4b** internal arc classification (IAC AFLR 40 kA 1s). It allows students and engineers to rehearse critical switching operations, racked withdrawal, key interlock sequences, and Lockout-Tagout (LOTO) procedures without physical hazard.

1.1 The 5-Bay Switchboard Lineup

Cubicle ID	Designation	Voltage / Bus	Equipment Type	Description & Operational Function
C1 · DG1	Diesel Generator 1	6.6 kV · Bus A	Incomer VCB (1250 A)	Primary power source: 3.2 MW medium-speed marine diesel generator delivering power to Bus Section A. Equipped with reverse-power and overcurrent relays.
C2 · BT1	Bus-Tie Breaker	6.6 kV · Bus A / B	Tie VCB (1250 A)	Sectionalizing interconnector coupling Bus A and Bus B. Allows split-bus redundant operation for DP compliance or parallel common-bus running.
C3 · BT-M1	Bow Thruster Motor	6.6 kV · Bus B	Feeder VCB (630 A)	High-power consumer: 1.8 MW tunnel thruster motor for vessel station-keeping and harbor maneuvering. Features integral EK6 earth switch and CT metering.
C4 · TR1-HV	Service XFMR 1 (HV)	6.6 kV · Bus B	Feeder VCB (630 A)	Step-down feeder supplying the 6.6 / 0.44 kV 1.5 MVA ship service distribution transformer. Key-interlocked with the secondary 440 V breaker.
C5 · ACB-LV	Service XFMR 1 (LV)	440 V · Secondary	Air Breaker (2000 A)	Low-voltage secondary incomer from TR1. Feeds the 440 V auxiliary switchboard for steering gear, pumps, lighting, and hotel loads.

2. Simulator Interface & Navigation Architecture

The simulator utilizes a purpose-designed **Industrial CAD / SCADA Interface** with soft industrial tones (#f4f6f8 background, #1f6f78 teal accents, #ad4337 high-voltage red warnings) designed to eliminate AI-clutter while maintaining maximum pedagogical clarity.

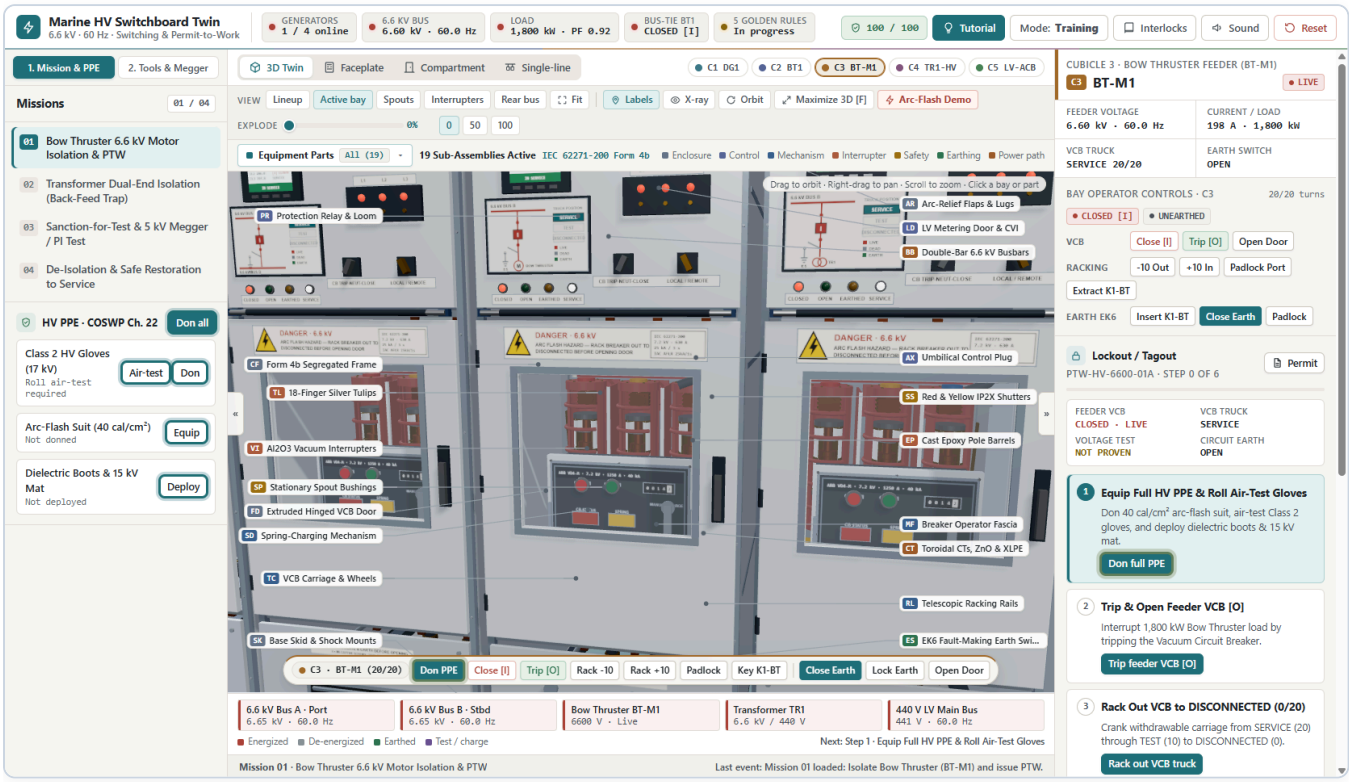


Figure 2.1 — Active Cubicle Focus (C3 · BT-M1) showing live REF615 relay HMI, 3D analog meters, single-line mimic diagram, and SCADA telemetry.

2.1 The Three-Panel Workspace

1. LEFT CONTROL DOCK (TWO TABS)

- **Tab 1 · Mission & PPE:** the four missions with step progress, and the HV PPE locker (arc-flash suit, air-tested Class 2 gloves, dielectric boots and 15 kV mat, or *Don all*).
- **Tab 2 · Tools & Megger:** the toolbelt (hand, 6.6 kV proving wand, portable earth), Castell key tracker, safety key box, and the 5 kV insulation / PI Megger. The tab opens automatically when the current step needs it.
- **Collapse chevron («):** hides the whole dock to give the 3D view more room.

2. RIGHT CONTROL DOCK (INSPECTOR & LOTO)

- **Bay Inspector:** live voltage, current, truck position and earth state for the selected cubicle, plus Bay Operator Controls (Trip/Close, Racking, Padlock, Castell keys, Earth EK6, Door).
- **Lockout / Tagout & Mission Steps:** the STCW / COSWP permit checklist. In Training mode the button for the current step pulses; in Exam / CoC mode no hints are given.
- **Collapse chevron (»):** hides the inspector when presenting.

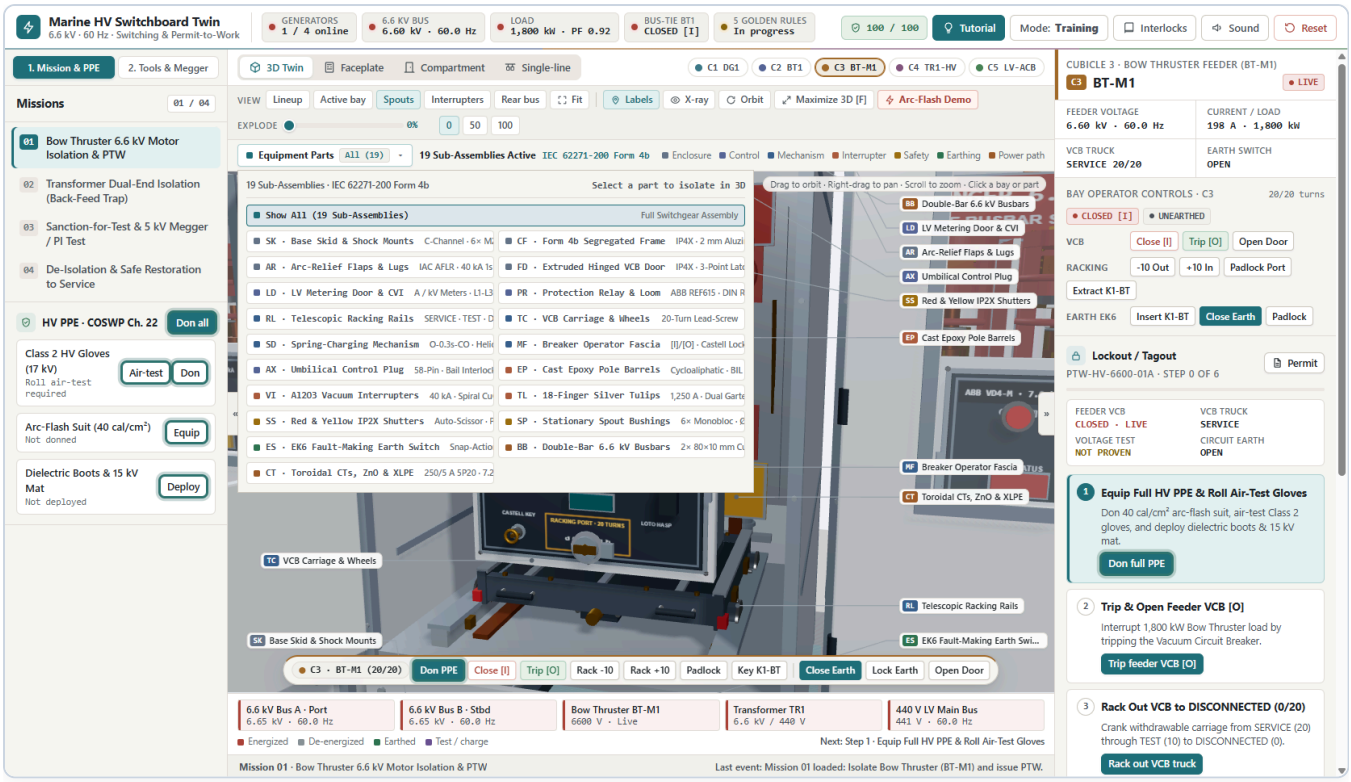


Figure 2.2 - The **Equipment Parts** dropdown lists all 19 sub-assemblies with category colours, so any part can be isolated with one click.

2.2 Center Viewport & 3D Navigation Controls

Control Action	Mouse / Desktop	Function & Cinematic Behavior
3D Orbit / Rotate	Left Mouse Button + Drag	Rotates camera smoothly around the active target point with inertia damping.
Pan Camera	Right Mouse Button + Drag	Translates camera horizontally and vertically across the switchboard face.
Zoom In / Out	Mouse Scroll Wheel	Moves camera closer to inspect terminal screws, springs, or labels.
Preset: Lineup	Click `Lineup` Button	Pulls back to a wide 5.3-meter vantage showing all 5 cubicles simultaneously.
Preset: Focus	Click `Focus` Button	Centers camera directly in front of the active cubicle at eye level (1.5 m).
Preset: Interior	Click `Interior` Button	Positions camera inside the breaker cell when front door is swung open.
Preset: Vacuum	Click `Vacuum` Button	Zooms into the cutaway vacuum interrupter bottle and contact gap.
Preset: Rear	Click `Rear` Button	Orbits behind the switchgear to inspect copper busbars and XLPE cable tails.
Exploded View Slider	Drag Slider (0% to 100%)	Smoothly drives all 19 sub-assemblies along collision-free explosion vectors.

Control Action	Mouse / Desktop	Function & Cinematic Behavior
Subsystem Isolation	Click Part Chip (e.g. `VI`)	Isolates selected sub-assembly while dimming surrounding parts to 14% ghosting.

3. Switchboard Cubicle Anatomy & Functional Details

In compliance with **IEC 62271-200 Form 4b**, each high-voltage cubicle is segregated by grounded earthed sheet-steel partitions into four isolated, self-contained compartments: the **Low-Voltage Metering Compartment** (top), the **Circuit Breaker Compartment** (center), the **Busbar Compartment** (upper rear), and the **Cable & Earthing Compartment** (lower rear).

3.1 Front Panel Instrumentation & Marine Features

1. ANALOG QUADRANT METERS (MOVING-IRON)

Dual 96×96 mm square analog quadrant meters with real **3D pivoting needles** deflected by live simulator state. The Voltmeter displays bus voltage (0–7.2 kV scale with red danger threshold above 6.6 kV). The Ammeter displays phase current (0–300 A scale) driven by motor/transformer load.

2. LIVE DOOR MIMIC & PILOT LAMPS

Screen-printed single-line mimic with live status indicators. A 4-lamp bank provides immediate status:

- **RED (CLOSED):** Breaker main contacts closed.
- **GREEN (OPEN):** Breaker open, circuit de-energized.
- **AMBER (EARTHED):** Earth switch closed to hull.
- **WHITE (SERVICE):** Truck fully racked in.

3. CAPACITIVE VOLTAGE INDICATOR (CVI)

Three neon flashers (L1, L2, L3) connected to capacitive tap rings on the monobloc spout bushings. LEDs illuminate brightly whenever conductors exceed 10% nominal voltage. Test jacks allow phase concordance testing.

4. NUMERICAL PROTECTION RELAY HMI (ABB REF615)

The door-mounted REF615 IED features a high-contrast backlit LCD displaying real-time 3-phase currents (IL1, IL2, IL3), operating voltages, and protection trip status. It integrates thermal overload (49), instantaneous overcurrent (50), time-delay overcurrent (51), and earth-fault (51N) curves.

5. MARINE STORM HANDRAIL & LOUVERS

As seen in commercial marine installations, a continuous tubular **stainless-steel storm handrail** with dark navy mounting brackets spans the horizontal seam of each bay. This allows marine engineers to safely brace themselves during heavy rolling in rough sea states while operating switches.

6. CABLE COMPARTMENT & LOUVERED DECOMPRESSION

The lower cover features stamped louvers over a dark recessed vent aperture. Four captive M6 hex screws secure the plate, enforcing the requirement that tools are mandatory to access terminal connections.

4. Circuit Breaker Withdrawable Cassette & Racking Mechanism

The marine vacuum circuit breaker (ABB VD4-M) is mounted on a withdrawable truck chassis. To achieve total electrical isolation, the breaker must be racked out mechanically along telescopic guide rails via an Acme lead-screw.

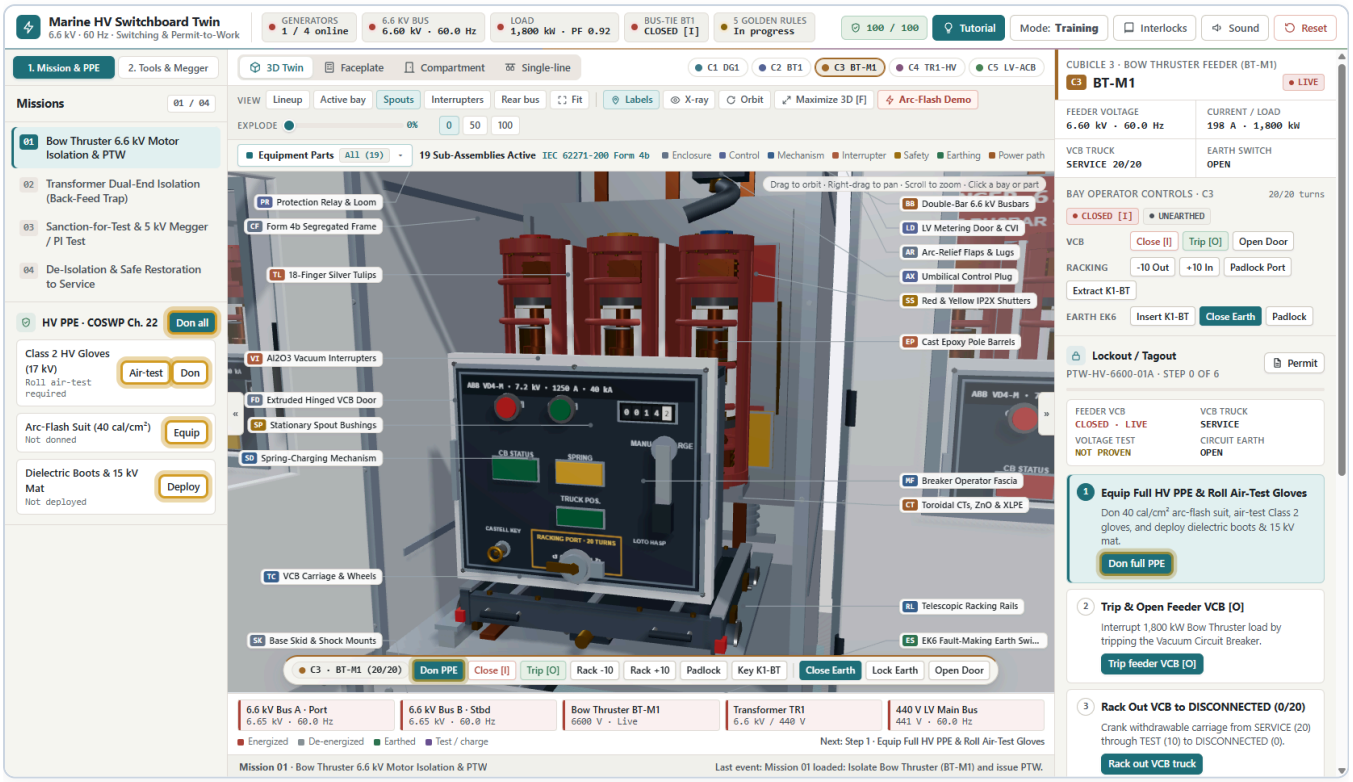


Figure 4.1 — VCB Withdrawn to DISCONNECTED position inside the cell, showing the Acme lead-screw, flanged carriage wheels, and automatic safety shutters.

4.1 The Three Operating Positions

Cassette Position	Crank Turns	Primary Contacts	Control Umbilical	Shutters & Safety Status
SERVICE	20 / 20 turns	Fully Plugged In	Connected	Shutters held open by carriage cam; primary tulip clusters engaged with stationary spout copper pins; cell door mechanically locked shut.
TEST	10 / 20 turns	Isolated (7 mm gap)	Connected	Primary power disconnected; secondary 58-pin control plug remains engaged; permits full electrical opening and closing tests without energizing load.
DISCONNECTED	0 / 20 turns	Isolated (+100 mm)	Disconnected	Shutters 100% closed and locked; primary contacts retracted behind IP2X partition; door interlock released allowing front door to be opened safely.

4.2 Safety Interlocks & Mechanical Shutter Flap

The racking mechanism features strict mechanical interlocks designed to prevent hazardous operator errors:

- **Closed-Breaker Racking Interlock:** A spring-loaded mechanical shutter flap physically obscures the racking hex socket whenever the circuit breaker contacts are closed. The breaker MUST be tripped open ([O]) before the racking crank can be inserted.
- **Door Interlock:** An espagnolette rod and detent bolt prevent the front door from opening while the breaker is in the SERVICE or TEST positions. Racking is carried out completely **behind the closed door**, with +1.8 cm positive internal clearance preventing any clipping.
- **Safety Shutters (SS):** Automatic red (Busbar spouts) and yellow (Cable spouts) metal barriers drop into place via scissor linkages as the carriage withdraws, preventing accidental touch contact with live 6.6 kV pins.

5. Complete 19 Sub-Assemblies Breakdown (100% Exploded View)

When the explosion slider is moved to 100%, the entire switchgear structure smoothly expands along precision CAD explosion vectors. All 19 sub-assemblies maintain a strict **minimum air clearance of >= 30 cm** with zero mutual collision, and the two-tier studio plinth dynamically expands to maintain floor clearance (minY >= 0.000 m).

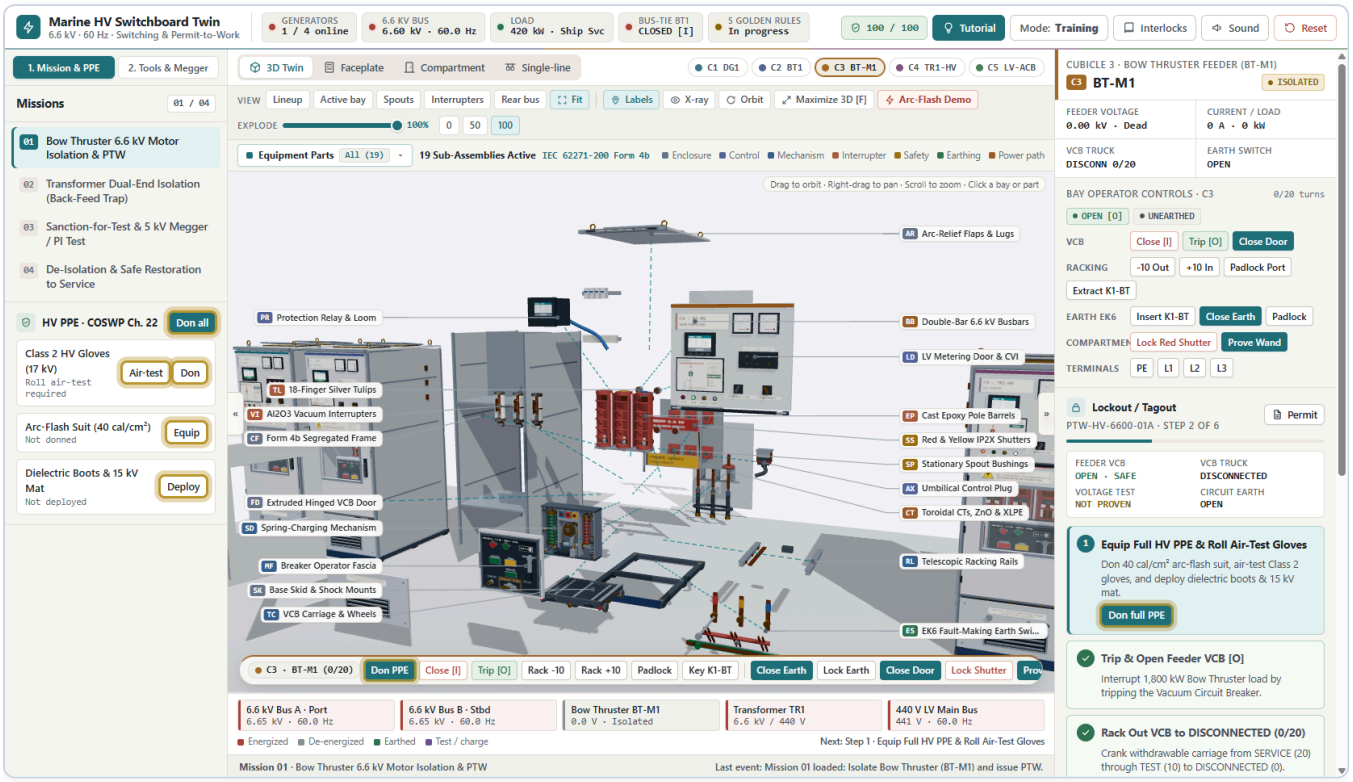


Figure 5.1 — 100% Exploded CAD View of Cubicle C3 (BT-M1) displaying all 19 segregated sub-assemblies with interactive callout badges.

5.1 Master Sub-Assembly Registry (IEC 62271-200 Form 4b)

Key	Sub-Assembly Name	Engineering Spec	Description & Construction Details
SK	Base Skid & Shock Mounts	C-Channel · 6× M20 Chocks	Welded structural steel C-channel base with elastomeric anti-vibration mounts, hold-down bolts, and navy kick-plates.
CF	Form 4b Segregated Frame	IP4X · 2 mm Aluzinc Steel	Modular sheet-steel frame with bolted upright posts, rivets, and earthed metallic partition decks between chambers.
AR	Arc-Relief Flaps & Lugs	IAC AFLR · 40 kA 1s	Top overpressure blowout damper flaps with perforated bronze flame-trap screens and 4 forged crane lifting eyebolts.
FD	Extruded Hinged VCB Door	IP4X · 3-Point Espagnolette	Extruded breaker compartment door with hollow bolted window frame, polycarbonate glass, latch rod, and hazard plates.

Key	Sub-Assembly Name	Engineering Spec	Description & Construction Details
LD	LV Metering Door & CVI	A / kV Meters · L1-L3 CVI	Upper door carrying moving-iron meters, mimic panel, 4-lamp pilot bank, CVI module, cam switches, and storm handrail.
PR	Protection Relay & Loom	ABB REF615 · DIN Rails	Numerical IED chassis, slotted DIN terminal rails, 18 screw-clamp blocks, 4 control MCBs, and armored wiring loom.
RL	Telescopic Guide Rails	Service · Test · Disconn	Machined C-section slide tracks with position detents, sliding copper grounding strip, and Acme lead-screw anchor.
TC	VCB Carriage & Wheels	20-Turn Acme Spindle	Withdrawable steel truck chassis with 4 concentric flanged wheels, central lead-screw nut, and frame earthing shoe.
SD	Spring-Charging Mechanism	O-0.3s-CO · Helical Coils	Stored-energy operating mechanism with helical close/trip springs, eccentric cam disc, geared charging motor, and coils.
MF	Breaker Operator Escutcheon	[I]/[O] · Castell Lock	Front fascia plate, Close/Trip pushbuttons, mechanical status flags, 5-digit cycle counter, manual lever, and shutter lock.
AX	Umbilical Control Plug	58-Pin · Bail Interlock	Heavy-duty multipin auxiliary connector with mechanical bail locking levers and flexible metallic control conduit.
EP	Cast Epoxy Pole Barrels	Cycloaliphatic · BIL 60 kV	Three phase-segregated cast resin pole housings with anti-creepage shed fins (210° cutaway revealing vacuum bottles).
VI	Vacuum Interrupters	40 kA · Spiral CuCr Contacts	Alumina ceramic vacuum bottles ($<10^{-4}$ Pa), spiral CuCr contacts, vapor arc shield, and 6-convolute stainless bellows.
TL	18-Finger Tulip Contacts	1250 A · Dual Garter Springs	Six circular self-aligning 18-finger silver-plated primary tulip clusters compressed by dual stainless steel garter springs.
SS	IP2X Safety Shutters	Auto-Scissor · Padlockable	Cam-actuated steel shutters (Red = Busbar, Yellow = Cable) with scissor linkages and lockout padlock hasp.
SP	Stationary Spout Bushings	6× Monobloc · Ø48 mm Pin	Rear bulkhead cast monobloc spouts with capacitive tap rings and silver-tipped stationary copper contact pins.
ES	EK6 Fault-Making Earth Switch	Snap-Action · Castell Keyed	Spring-assisted twin-blade copper earthing knives, ship hull PE copper busbar, Castell key lock, and CME terminals.
BB	Double-Bar 6.6 kV Busbars	2× 80×10 mm Cu · L1/L2/L3	Electrolytic copper main busbars with phase heat-shrink sleeves (Red/Yellow/Blue), splice joints, and shedded standoffs.
CT	Toroidal CTs, ZnO & XLPE	250/5 A 5P20 · 7.2 kV ZnO	Cast toroidal protection CTs, core-balance CT, ZnO surge arresters, elastomer stress cones, and armored 6.6 kV XLPE cables.

6. Vacuum Arc Interruption & Contact Physics

The vacuum circuit breaker relies on the extraordinary dielectric strength of a high vacuum ($< 10^{-4}$ Pa). When contacts separate under load, current vaporizes a minute quantity of contact metal, creating a metal vapor arc that burns across the gap until current zero.

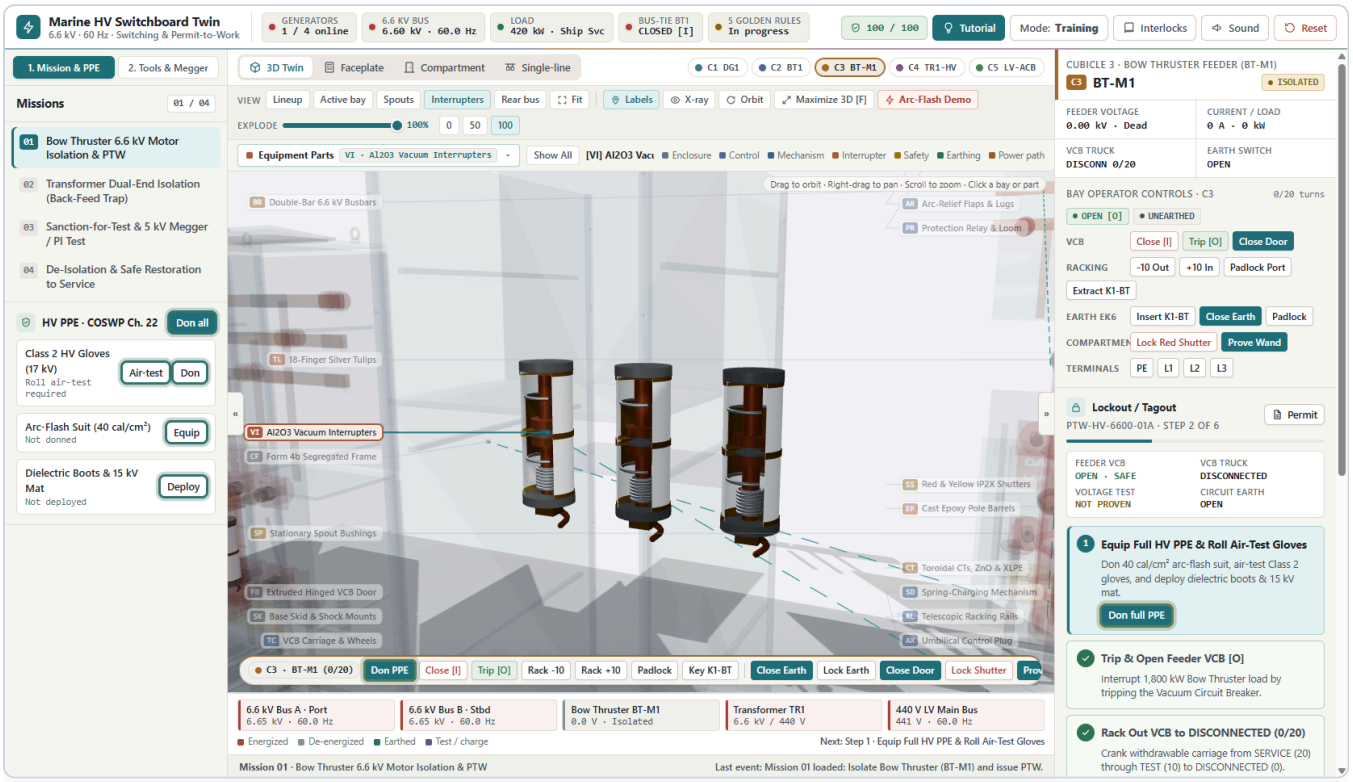


Figure 6.1 — High-Resolution Cutaway of the Alumina Ceramic Vacuum Interrupter (VI), revealing the spiral CuCr contact geometry, arc condensation shield, and flexible bellows.

6.1 Contact Metallurgy & Transverse Magnetic Field (TMF)

Standard flat contacts would suffer severe localized anode spots and melting at 40 kA fault levels. The simulator faithfully models **Spiral-Slotted Copper-Chromium (CuCr 75/25)** contact discs:

- **Arc Rotation:** The spiral geometry forces fault current through inclined path segments, producing a powerful self-generated magnetic field transverse to the arc column.
- **Diffuse Arc Mode:** The Lorentz force drives the arc rapidly around the circumference of the contact surface at speeds exceeding 100 m/s, preventing localized overheating and electrode erosion.
- **Condensation Shield:** An internal cylindrical bronze vapor shield intercepts evaporated metal droplets, preventing metallic deposition on the inner ceramic envelope wall that would degrade insulation resistance.
- **Kinematic Stroke:** The moving stem is supported by a 6-convolute stainless steel bellows allowing a **12 mm contact travel** in under 45 milliseconds during tripping.

7. Secondary Distribution: 440 V LV Air Circuit Breaker (C5)

Cubicle 5 represents the Low-Voltage (LV) secondary incomer from Service Transformer 1. While operating at 440 V, it presents extreme prospective short-circuit fault current levels (up to 65 kA) due to low transformer impedance.

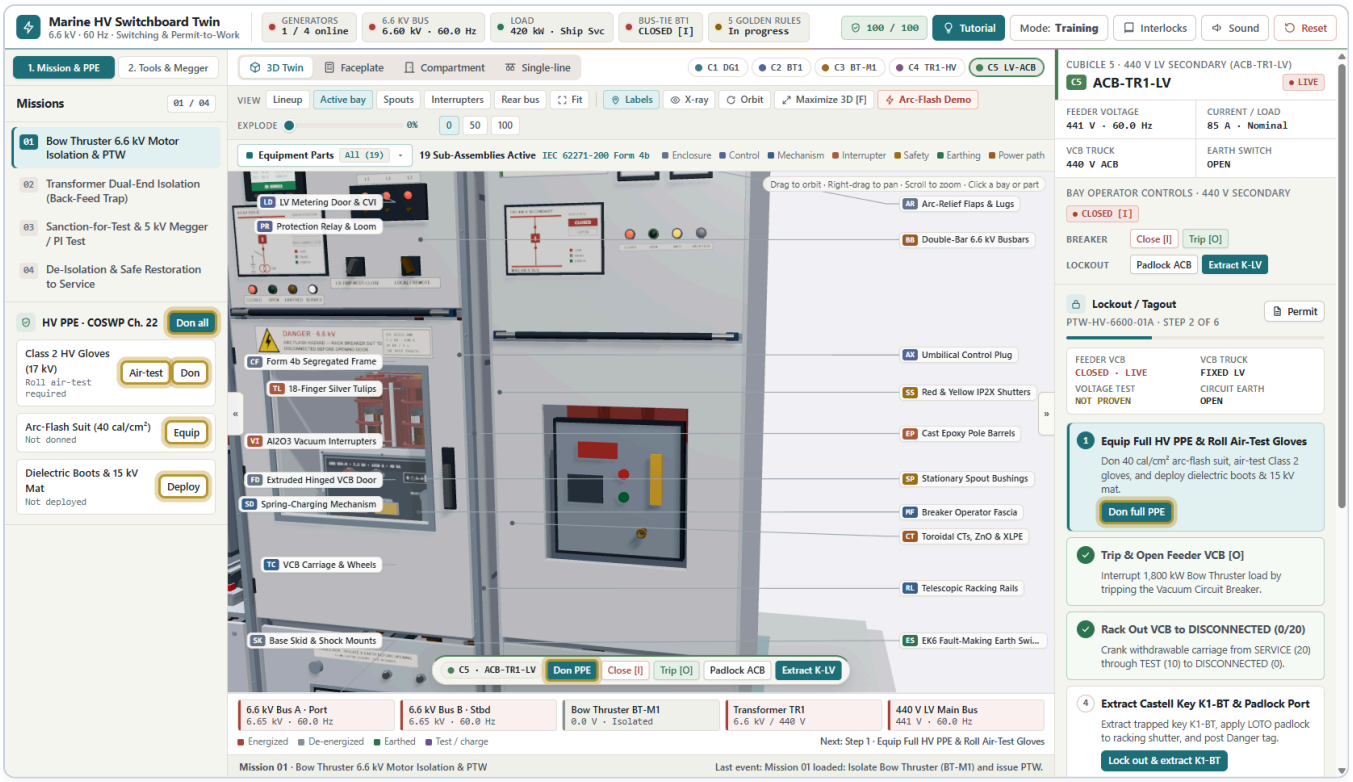


Figure 7.1 — Cubicle C5 (ACB-TR1-LV) showing the ABB Emax 2 air circuit breaker chassis, de-ionizing arc chute splitters, live 440 V meters, and tie bus duct.

7.1 LV Breaker Construction vs. HV Breaker

1. AIR CIRCUIT BREAKER (ABB EMAX 2)

Unlike the vacuum interrupters of C1–C4, C5 utilizes atmospheric air arc extinction. Above each phase contact sits a heavy **De-Ion Arc Chute** containing 7 slotted steel splitter plates that stretch, cool, and magnetically divide the arc into series sub-arcs, driving arc voltage above system voltage to force extinction.

2. TRANSFORMER BACK-FEED HAZARD & KEY K-LV

If 440 V auxiliary generators are online, energizing C5 can step voltage back up through TR1 to 6.6 kV on C4, creating a deadly back-feed. A mechanical **Castell Key (K-LV)** enforces that C5 must be locked OPEN before C4 can be safely isolated and earthed.

8. STCW & COSWP Marine LOTO Switching Procedures

The simulator enforces strict adherence to the **Code of Safe Working Practices for Merchant Seafarers (COSWP Chapter 16)** and STCW High-Voltage training directives. Every switching operation follows the universally recognized **Five Golden Safety Rules**.

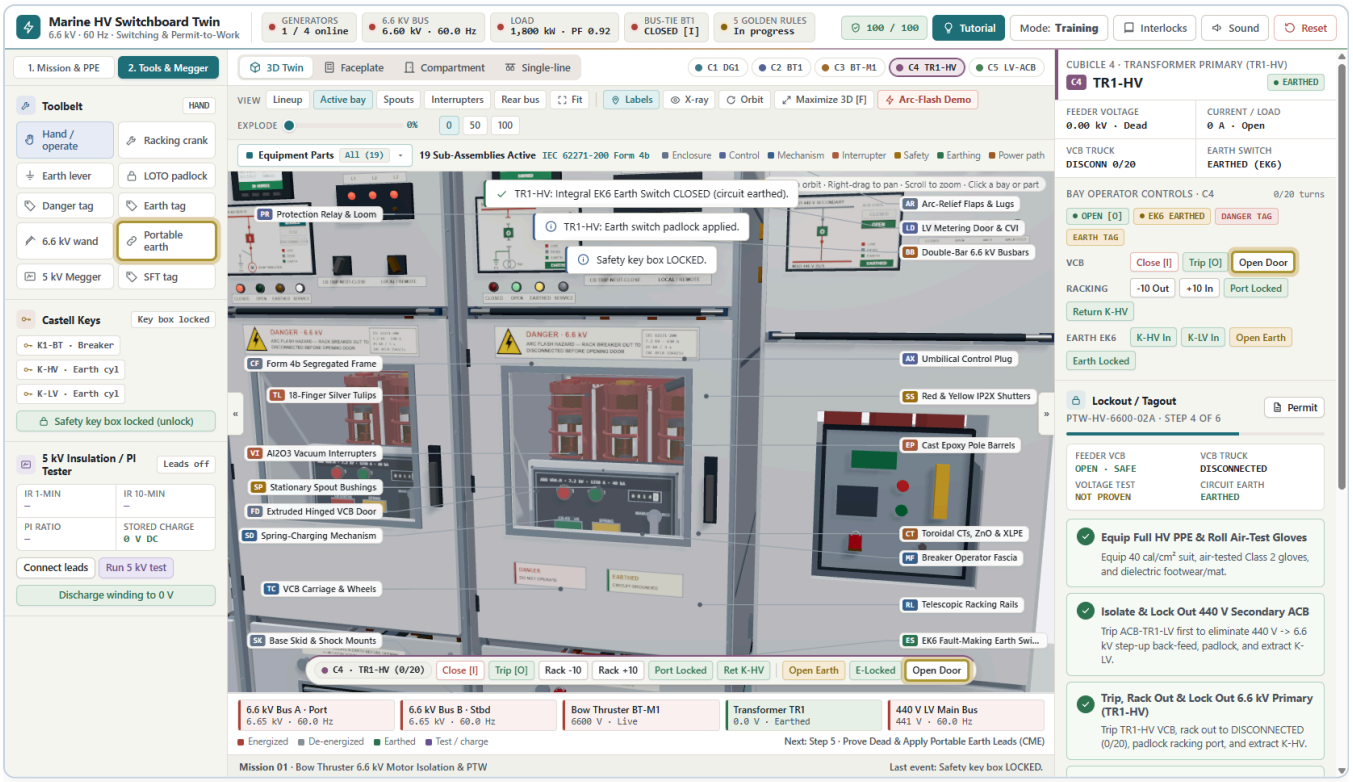


Figure 8.1 — Interactive LOTO Permit Panel and Interlock Matrix guiding the step-by-step safe isolation and earthing of Transformer 1 (Mission 2).

8.1 The 4 Built-In Training Missions

Mission ID	Title / Scope	Target Cubicle	Key Learning Objectives & Mandatory Safety Steps
Mission 1	Bow Thruster Motor Planned Maintenance Isolation	C3 · BT-M1 (6.6 kV)	Stop motor machinery -> Trip VCB open -> Rack breaker to DISCONNECTED -> Lockout racking shutter -> Verify dead -> Close EK6 earth switch -> Lock earth switch -> Apply LOTO warning tags.
Mission 2	Transformer HV / LV Dual-Key Isolation	C4 · TR1-HV & C5 · ACB-LV	Open LV ACB C5 to eliminate back-feed -> Remove Castell Key K-LV -> Transfer key to HV bay C4 -> Rack out C4 VCB -> Close HV earth switch -> Extract Permit-to-Work release key.
Mission 3	Arc-Fault Blast Event & Emergency Response	C1 · DG1 (6.6 kV)	Simulated arc blast trip -> Emergency trip bus-tie BT1 -> Inspect arc-relief dampers (AR) -> Don full flash suit -> Check gas

Mission ID	Title / Scope	Target Cubicle	Key Learning Objectives & Mandatory Safety Steps
			venting -> Execute safe isolation under emergency protocols.
Mission 4	Insulation Resistance Megger Testing & SFT	C3 · BT-M1 (Motor Feeder)	Issue Sanction-for-Test (SFT) -> Don 15 kV boots/gloves -> Open earth switch -> Connect 5 kV Megger test leads to L1 & PE -> Conduct 1-minute 5000 V test -> Prove > 1000 MΩ -> Discharge cable.

9. Live-Line Voltage Proving & Earthing Execution

Conductors must never be assumed dead until physically proven de-energized with an approved high-voltage live-line tester.

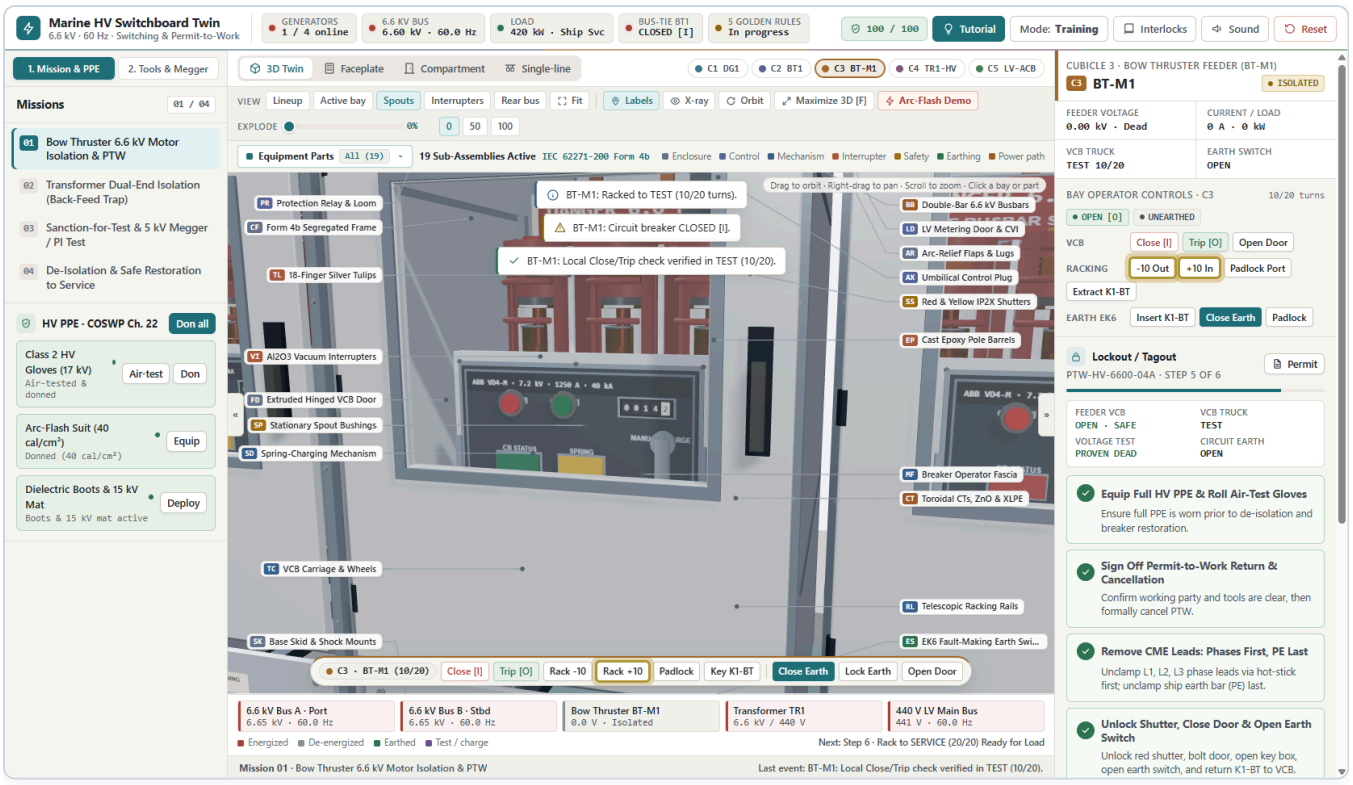


Figure 9.1 — 3D Telescopic 6.6 kV Live-Line Proving Wand conducting the mandatory 3-point test on feeder terminal L2 prior to earthing.

9.1 The Mandatory 3-Point Test Rule ("Prove - Test - Prove")

1. **Prove Instrument:** Insert the proving wand tip into an approved portable piezoelectric or battery HV proving unit (or CVI test socket). Confirm bright visual LED flashing and audible buzzer sounding.
2. **Test Conductors:** Apply the proving wand tip to each phase terminal (L1, L2, L3) inside the switchboard cable compartment. Verify complete absence of light and sound.
3. **Re-Prove Instrument:** Re-test the wand immediately on the proving unit to confirm the instrument did not suffer internal circuit failure or dead battery during the test.

9.2 Application of Portable Earth Leads (CME)

> **Mandatory Earthing Rule:** Always attach the grounding clamp to the vessel's hull Protective Earth (PE) copper bar **FIRST**. Only then apply the phase clamps to conductors L1, L2, and L3 using an insulated fiberglass operating stick. When removing earths, strictly reverse the order: remove phase clamps first, ship hull PE clamp last!

10. Safety Interlock Matrix & Troubleshooting Reference

Interlock Condition	Physical / Electrical Mechanism	Corrective Operator Procedure
Racking handle cannot be inserted into VCB port	Mechanical racking shutter flap is engaged because circuit breaker contacts are CLOSED [I].	Press green Trip pushbutton [O] on the breaker escutcheon or trip via SCADA before inserting the racking crank.
Breaker compartment door will not open	Door interlock bolt blocked because breaker carriage is in SERVICE or TEST position.	Crank breaker counter-clockwise 20 full turns until the mechanical position flag indicates DISCONNECTED (0/20).
Earth switch cannot be closed	Mechanical linkage interlocked with VCB position. Prevents earthing a breaker that is racked in.	Ensure breaker is fully in DISCONNECTED position and padlocked against racking in before attempting earth switch operation.
Earth switch operating socket blocked	Castell key missing from front cylinder (e.g., C4 requires Key K-LV from secondary bay C5).	Open the secondary LV ACB on C5, extract Key K-LV, insert into C4 earth switch Castell lock, and turn 90° clockwise.
Cannot close VCB	Earth switch is closed, or Castell key locked out, or closing spring discharged.	Open and padlock earth switch, charge closing spring manually or via motor, and release Castell key interlock.

Emergency Electrical Shock Protocol: In the event of an electrical contact casualty, do NOT touch the victim with bare hands. Immediately press the nearest emergency stop (ESD) or trip incoming generators. If power cannot be immediately severed, use the non-conductive rescue hook to pull the casualty clear before commencing CPR and defibrillation.

11. Scoring, Arc-Flash Hazard System & Presentation Tools

11.1 Two-Tier Violation System (every mistake = -5 points)

Every wrong action deducts exactly **5 points** from the 100-point assessment score, but the simulator reacts differently depending on how dangerous the mistake would be on a real ship.

Tier	Typical triggers	Feedback the student gets
Tier 1 · Mechanical / procedural block	Racking a closed VCB, key not in hand, padlocked port, incomplete PPE, door interlocked, wrong earthing order.	Amber warning toast with the rule broken, a metallic clunk and double buzzer, score badge turns amber/red. Exam mode also shows a full notice dialog.
Tier 2 · Live electrical / arc-flash hazard	Applying portable earth to a live terminal, probing or megger-testing a live 6.6 kV feeder, cancelling a Sanction-for-Test before discharging 4,850 V DC.	Full 3D plasma blast in the active bay, loud blast sound with ECR klaxon, and an ISO 7010 W012 / NFPA 70E hazard banner (40 cal/cm², 6,600 V, 40 kA). The view switches to 3D automatically.

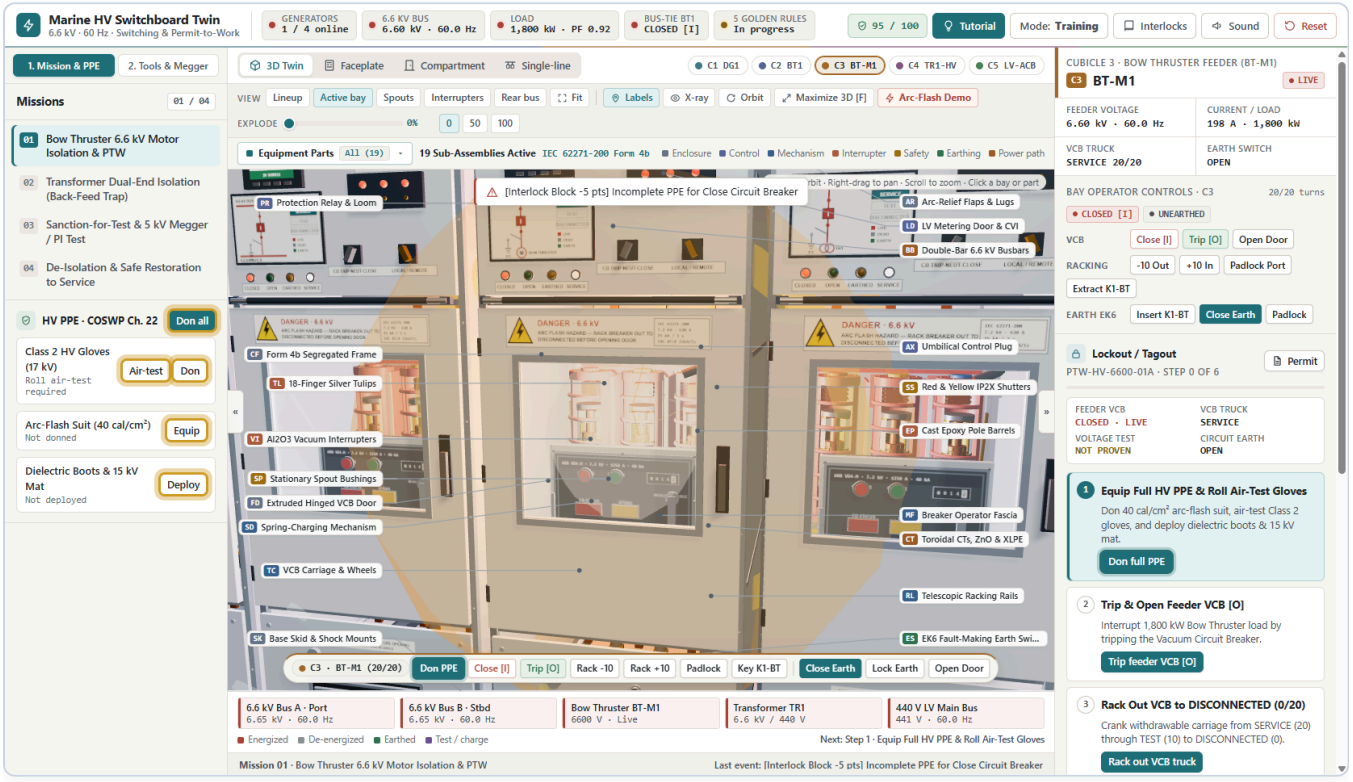


Figure 11.1 - A Tier 1 interlock block: the breaker cannot be closed without full PPE; the toast names the rule and the score drops by 5.

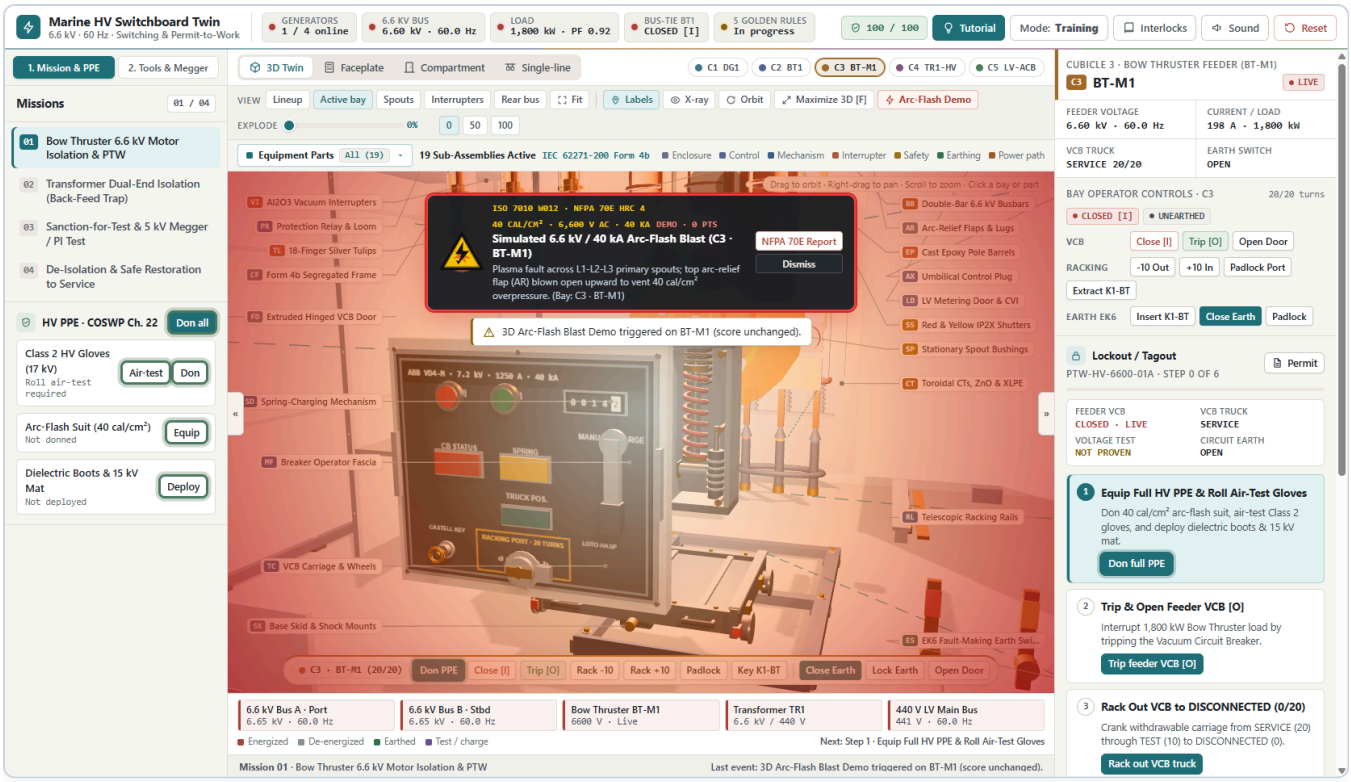


Figure 11.2 - A Tier 2 arc-flash event: plasma flash and lightning across the primary spouts, arc-relief flap venting upward, red screen-edge vignette and the ISO 7010 / NFPA 70E hazard banner.

> **Classroom demo:** click `Arc-Flash Demo` in the 3D toolbar to trigger the full blast on demand. It deducts no points. Use *NFPA 70E Report* on the banner to open the incident-energy summary.

11.2 Theater 3D Mode & Quick-Action Bar

- **Maximize 3D (F):** collapses both sidebars so the switchboard fills the screen (ideal for projectors). Press the button again, or `F`, to restore the docks. The left and right chevrons collapse each dock individually.
- **Quick-Action Bar:** a floating bar at the bottom of the 3D view shows the selected bay and its main controls - Don PPE, Close, Trip, Rack -10 / +10, Padlock, Castell key, Earth, Lock Earth and Open/Close Door - so you can operate the switchboard without leaving the 3D view. Every button uses the same interlock checks as the right-hand inspector.



Figure 11.3 - Theater 3D mode with both docks collapsed and the Quick-Action Bar along the bottom edge.

11.3 Guided Training Aids

- **Step spotlight:** in Training mode the control needed for the current LOTO step glows with a teal/amber pulse. Switch to *Mode: Exam / CoC* to remove all hints for assessment.
- **Tools & Megger tab:** the left dock stays short enough to avoid scrolling. Mission 3 (Sanction-for-Test and 5 kV PI test) opens this tab automatically.

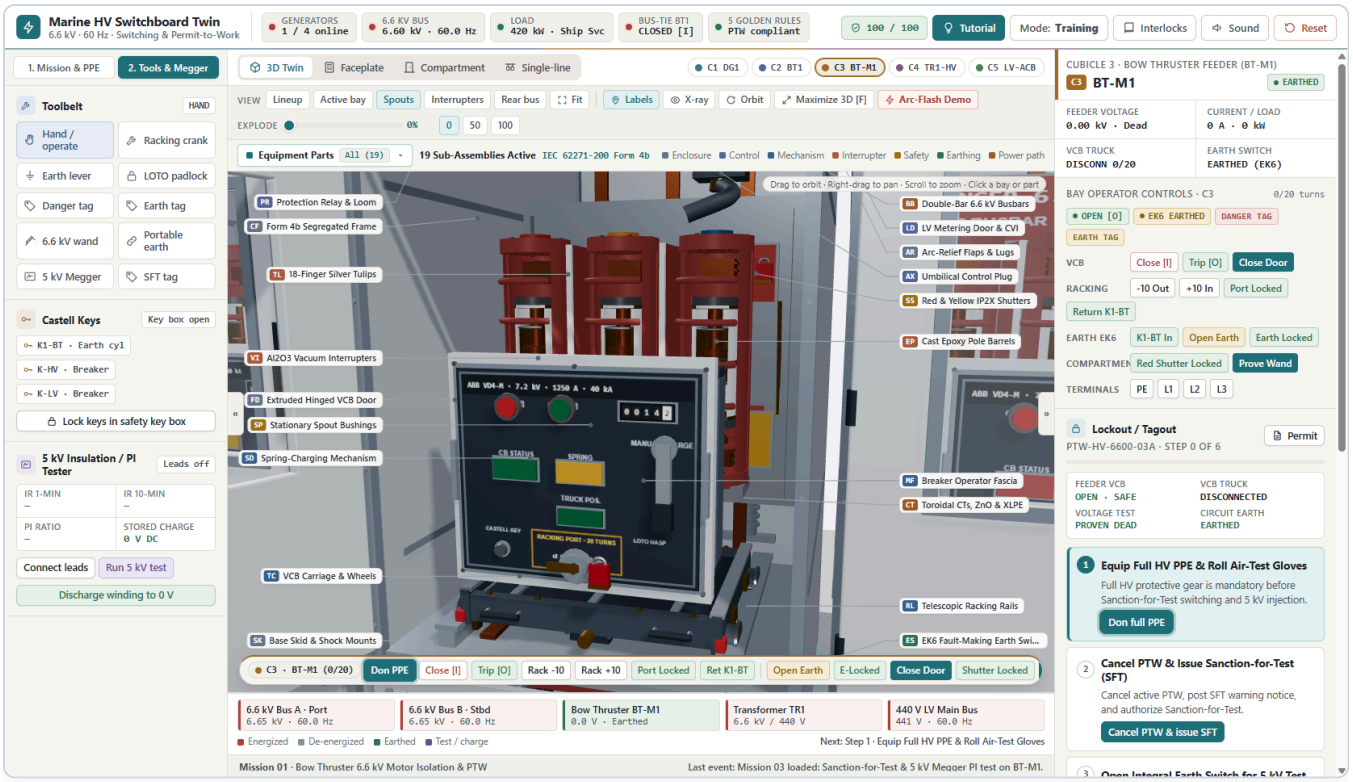


Figure 11.4 - The **2. Tools & Megger** tab showing the toolbelt, Castell key tracker, safety key box and 5 kV insulation tester.

11.4 Mechanism Close-up

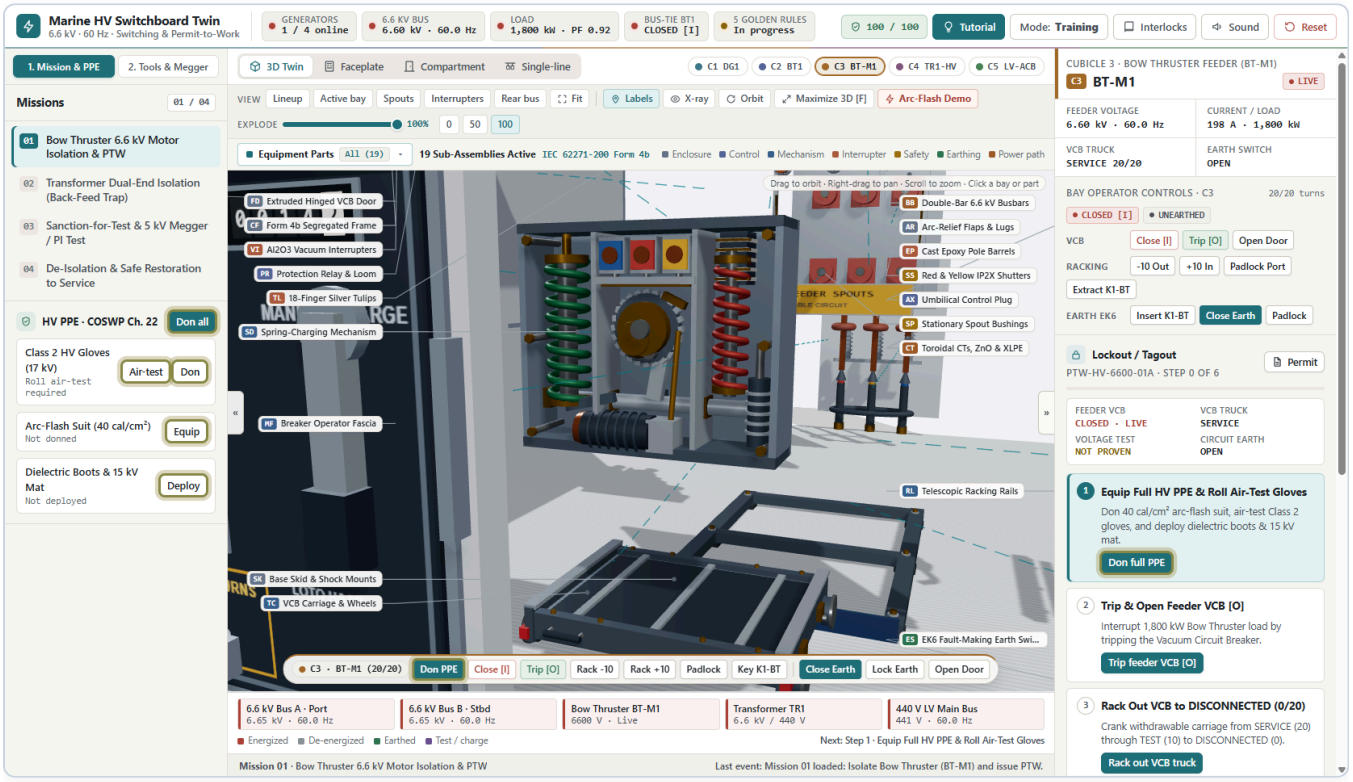


Figure 11.5 - Exploded VCB operating mechanism (carriage, spring-charging mechanism and fascia) at 100% explode.